Katugampola Fractional Derivative

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1 Introduction

The Katugampola fractional derivative is a generalization of the classical derivative, which is defined in Equation 1

$$D_a^\alpha f(x) = \frac{1}{\Gamma(m-\alpha)} \frac{d^m}{dx^m} \int_0^x (x-t)^{m-\alpha-1} f(t) dt, \text{ where } m = \lceil a \rceil \text{ and } \alpha \in (0,1) \tag{1}$$

2 Derivation of Katugampola Fractional Derivative

To derive Equation 1, we start with the Riemann-Liouville fractional integral as in Equation 2

$$I_{0+}^\alpha f(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt, \tag{2}$$

where $0 < \alpha < 1$. Then, using the classical Caputo fractional derivative, we have:

$$\begin{aligned} D_a^\alpha f(x) &= \frac{1}{\Gamma(m-\alpha)} \frac{d^m}{dx^m} I_{0+}^\alpha f(x) \\ &= \frac{1}{\Gamma(m-\alpha)} \frac{d^m}{dx^m} \left(\frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt \right) \end{aligned} \tag{3}$$

This leads to Equation 1

3 Property

3.1 Linearity

The Katugampola fractional derivative satisfies the following property of linearity:

$$D_a^\alpha (cf + g) = cD_a^\alpha f + D_a^\alpha g, \tag{4}$$

where c is a constant.

3.2 Product Rule

The product rule for the Katugampola fractional derivative is given by:

$$D_a^\alpha (fg) = D_a^\alpha f \cdot g + f \cdot D_a^\alpha g - \alpha D_a^\alpha f \cdot D_a^\alpha g \tag{5}$$

where f and g are differentiable functions.

4 Conclusion

The Katugampola fractional derivative provides a useful tool for generalizing the concept of differentiation. Its properties, such as linearity and the product rule, make it applicable in various mathematical contexts.

Equations (1), (4) and (5) illustrate key aspects of the Katugampola fractional derivative.